

2025 AMC 10B Problem 22 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 22

Problem:

A seven-digit positive integer is chosen at random. What is the probability that the number is divisible by 11, given that the sum of its digits is 61?

Answer Choices:

A. $\frac{3}{14}$

B. $\frac{3}{11}$

C. $\frac{2}{7}$

D. $\frac{4}{11}$

E. $\frac{3}{7}$

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Solution 1:

Note that the maximum digit sum of a 7-digit number is 63. To compute how many 7-digit numbers have a digit sum of 63, we may think about it as distributing two -1 's to the digits of 9999999. If they are distributed to the same digit, there are 7 ways to do so and if they are distributed to different digits there are $\binom{7}{2} = 21$ ways for a total of 28 numbers with digit sum 63.

A multiple of 11 must have an alternating digit sum of 11 as well, and note that 9999999 has an alternating digit sum of 9. Therefore, both -1 's must be distributed to digit places which are negative - the 2nd, 4th, and 6th slots. Similarly to above there are $3 + \binom{3}{2} = 6$ ways to do this for a final answer of

$$\frac{6}{28} = \boxed{(A) \frac{3}{14}}.$$

Solution 2:

Suppose the 7-digit number was $\overline{A_1 A_2 A_3 A_4 A_5 A_6 A_7}$. Then, let $A = a_1 + a_3 + a_5 + a_7$ and $B = a_2 + a_4 + a_6$. Note that $B \leq 9 \cdot 3 = 27$, and $A \leq 9 \cdot 4 = 36$. Thus, $34 = 61 - 27 \leq 61 - B = A \leq 36$, and $25 = 61 - 36 \leq 61 - A = B \leq 27$. Thus, $7 \leq A - B \leq 11$.

Note that if $\overline{A_1 A_2 A_3 A_4 A_5 A_6 A_7}$ is divisible by 11, $A - B$ must also be divisible by 11, so $A - B = 11$, implying $A = 36$ and $B = 25$. Note that $(9 - a_2) + (9 - a_4) + (9 - a_6) = 27 - B = 2$, so we can choose them in

$$\binom{2+3-1}{3-1} = \binom{4}{2} = 6$$

ways for a_2, a_4, a_6 . As $A = 36$, $a_1 = a_3 = a_5 = a_7 = 9$, so thus there are a total of 6 such 7-digit numbers divisible by 11 and sum 61.

To count the total number of 7-digit numbers with digit sum 61, we note that

$$\sum_{k=1}^7 (9 - a_k) = 63 - \sum_{k=1}^7 a_k = 2$$

so we can distribute them in

$$\binom{2+7-1}{7-1} = \binom{8}{6} = 28$$

ways, for a probability of

$$\frac{6}{28} = \boxed{\text{(A)} \frac{3}{14}}.$$

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